

MICRONUTRIENTS

Fat-Soluble Vitamins

VITAMIN A (RETINOL)

Etiology and Pathogenesis

Vitamin A is a fat-soluble vitamin essential for retinal photoreceptor function, epithelial cell proliferation, and keratinization. The two most clinically important metabolites are **retinal**, a key component in rhodopsin formation, and **retinoic acid**, which regulates cell differentiation.

Dietary sources of vitamin A include: - **Plant sources:** dark green leafy vegetables, red palm oil, and brightly colored fruits such as papaya, mango, carrots, tomatoes, apricots, and cantaloupe. In plants, vitamin A is present as β -carotene, which is converted to retinal in the body. - **Animal sources:** egg yolk, liver, fish, fortified milk, and other dairy products. In these sources, vitamin A exists as retinyl esters.

In the intestine: - β -carotene from plants is converted to retinal and then reduced to retinol in intestinal villous cells. - Retinyl esters from animal sources are hydrolyzed to retinol in the intestinal lumen and absorbed into mucosal cells. - All absorbed retinol is re-esterified to retinyl esters in the intestinal mucosa, packaged into chylomicrons, and transported to the liver for storage.

In the liver: - Vitamin A is stored as retinyl esters. - When needed, it is released as retinol bound to retinol-binding protein (RBP) and transthyretin for systemic circulation.

Vitamin A Deficiency

Epidemiology

Vitamin A deficiency (VAD) can lead to both cutaneous and ocular complications. It is the leading cause of **preventable childhood blindness** worldwide, according to the WHO. VAD is also associated with impaired immune function and increased susceptibility to infections.

Etiology and Pathogenesis

Primary causes of VAD include: - Inadequate dietary intake - Fat malabsorption syndromes - Liver disease

In the United States, risk factors include: - Eating disorders - Restrictive diets - Chronic illness

Due to its fat-soluble nature, vitamin A absorption depends on normal fat digestion. Conditions that impair fat absorption increase the risk of VAD, including: - Pancreatic insufficiency - Biliary tract disease - Celiac disease - Crohn disease - Shwachman-Diamond syndrome - Cystic fibrosis - Cholestatic liver disease - Gastric bypass surgery

Clinical Findings

Ocular manifestations are typically the earliest signs: - **Nyctalopia** (impaired dark adaptation) - **Xerophthalmia** (dry eyes) - **Bitot spots** – foamy, white patches on the conjunctiva due to keratin debris and *Xerosis bacillus* overgrowth - **Corneal xerosis**, ulceration, and **keratomalacia** (softening of the cornea), which can progress to corneal perforation, iris prolapse, and permanent blindness

Cutaneous manifestations result from abnormal keratinization: - Mild deficiency: **xerosis** and scaling - Severe deficiency: **dermomalacia** — deep, painful skin fissures

Essential Fatty Acid (EFA) Deficiency

Clinical Features

Affected individuals may exhibit: - Hyperpigmentation or hypopigmentation of the hair - Fatty liver infiltration - Increased susceptibility to infection - Blunted immune response - Anemia - Thrombocytopenia - Growth retardation

Laboratory Testing

In EFA deficiency: - **Linoleic acid** levels are low - The enzymes that convert linoleic acid to arachidonic acid instead use oleic acid, producing the abnormal metabolite **5,8,11-eicosatrienoic acid** - Laboratory findings include: - Decreased plasma levels of **linoleic acid** and **arachidonic acid** - Elevated plasma levels of **5,8,11-eicosatrienoic acid** - A plasma ratio of **5,8,11-eicosatrienoic acid to arachidonic acid ≥ 0.2** is diagnostic of EFA deficiency

Treatment

- Topical application of linoleic acid-rich oils (e.g., sunflower seed oil, safflower oil) may improve skin findings, but absorption is unpredictable
- Optimal treatment involves **oral or intravenous supplementation** of essential fatty acids
- To prevent deficiency, EFAs should constitute **1% to 2% of total daily caloric intake**